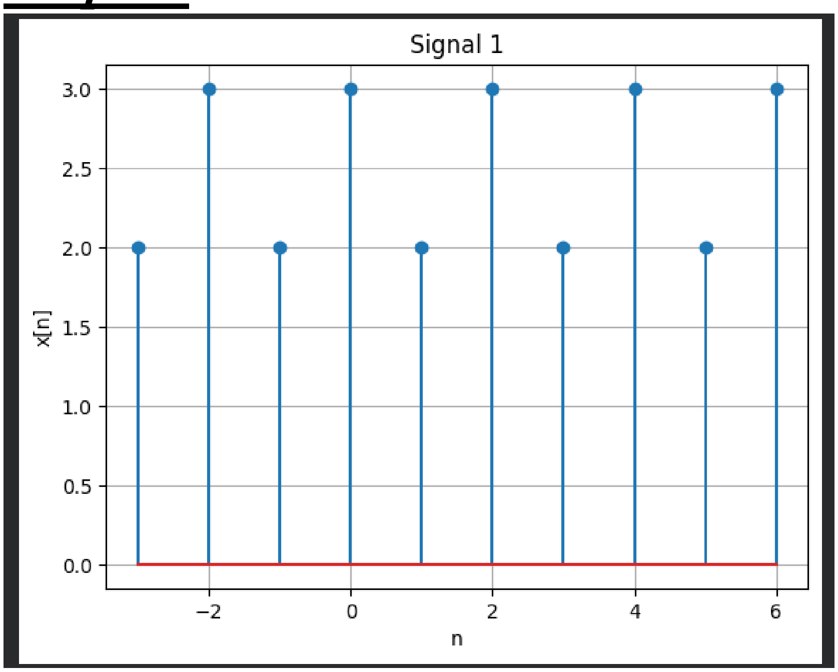
	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:** Open Ended Problem 1	
Open Ended Problem 1	**Date:15/7/2026**	**Enrollment No:92400133050**

Problem 1 :

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-3, 7)
x = [2,3,2,3,2,3,2,3,2,3]
plt.stem(n, x)
plt.title("Signal 1")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

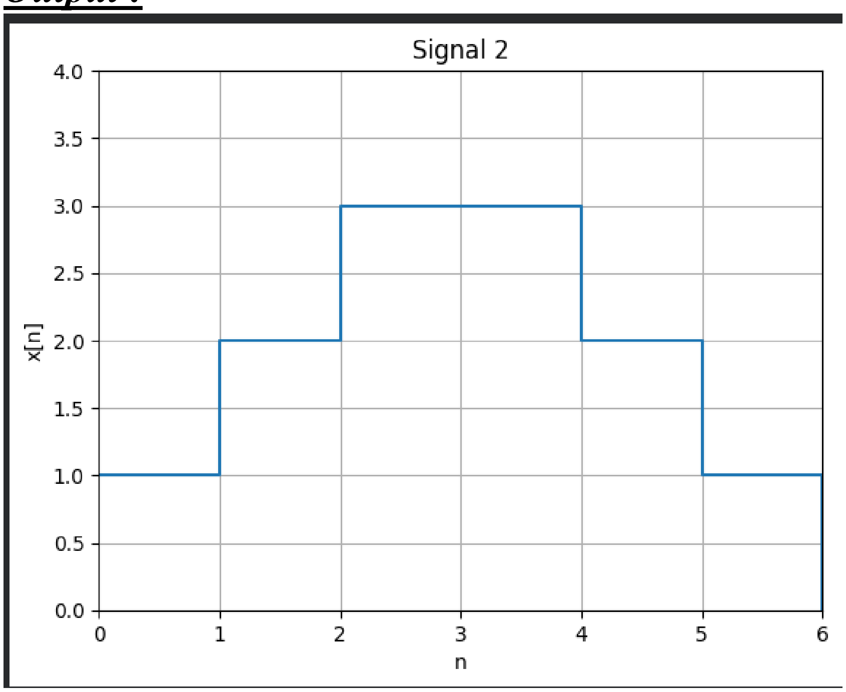
Output :



Problem 2 :

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0, 7)
x = [1,2,3,3,2,1,0]
plt.step(n, x, where='post')
plt.xlim(0,6)
plt.ylim(0,4)
plt.title("Signal 2")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

Output :



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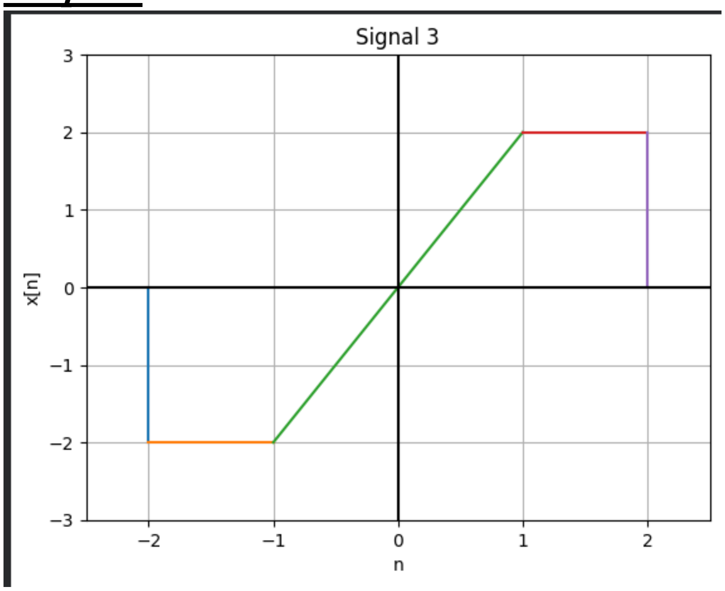
Problem 3 :

```
import numpy as np
import matplotlib.pyplot as plt
x1 = np.linspace(-2,-1,100)
y1 = -2*np.ones(100)
x2 = np.linspace(-1,1,100)
y2 = 2*x2
x3 = np.linspace(1,2,100)
y3 = 2*np.ones(100)
plt.plot([-2,-2],[0,-2])      # Vertical line at x = -2
plt.plot(x1,y1)
plt.plot(x2,y2)
plt.plot(x3,y3)
plt.plot([2,2],[2,0])        # Vertical line at x = 2
plt.axhline(0,color='black')
plt.axvline(0,color='black')
plt.xlim(-2.5,2.5)
plt.ylim(-3,3)
plt.grid(True)
plt.title("Signal 3")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.show()
```

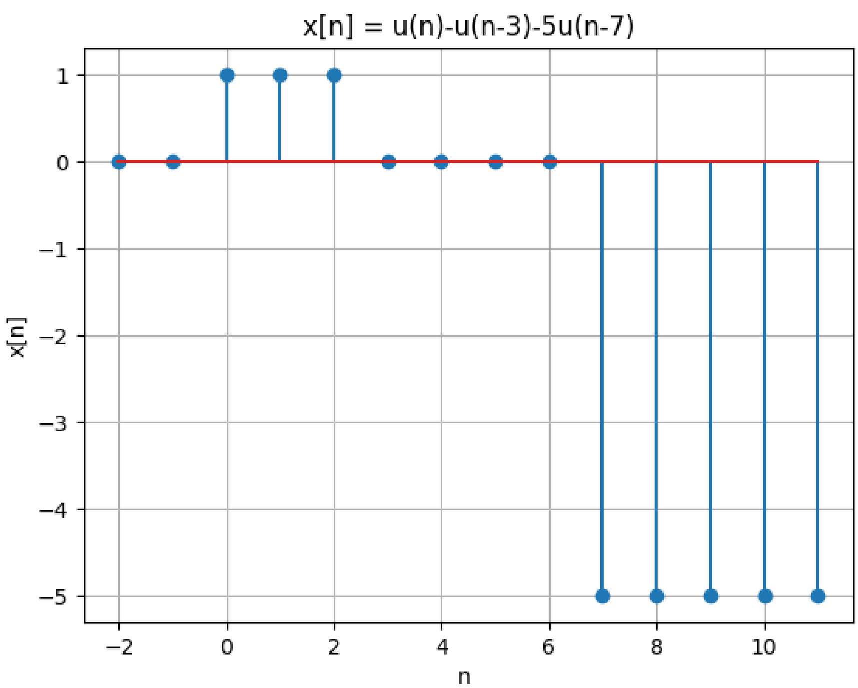
Output :



Problem 4:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-2, 12)
u = lambda x: np.where(x >= 0, 1, 0)
x = u(n) - u(n-3) - 5*u(n-7)
plt.stem(n, x)
plt.title("x[n] = u(n)-u(n-3)-5u(n-7)")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

Output :



Problem 5:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-3, 4)
delta = lambda x: np.where(x == 0, 1, 0)
x = delta(n) + 3 * delta(n-1) + 5 * delta(n+1)
plt.stem(n, x)
plt.title("x[n]=δ(n)+3δ(n-1)+5δ(n+1)")
plt.xlabel("n ")
plt.ylabel("x[n]")
plt.grid(True)
plt.show()
```

Output :